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RATUL **THAKUR**

Graduate Student at the International Centre for Theoretical Sciences, Bengaluru

Education

2023-present **DOCTOR OF PHILOSOPHY, THEORETICAL PHYSICS**
International Centre for Theoretical Sciences (ICTS-TIFR) · Bengaluru, India

2019-2023 **BACHELOR OF TECHNOLOGY, ENGINEERING PHYSICS**
Indian Institute of Technology · Roorkee, India
Bachelor's Thesis: Slave Rotor Mean Field Theory of Strongly Correlated Fermions: A study of Phase Transitions on the Hubbard Model.
Grade (CGPA): 8.4/10

2019 **HIGH SCHOOL**
St. Mary's School · Sundernagar, India
Grade: 96.6%

Research

INTERESTS

- **Quantum Many-Body theory**, focusing primarily on the dynamical aspects of closed many body systems. Quantum Chaos and thermalization (via the Eigenstate Thermalization Hypothesis) and the lack of thermalization in systems with strong disorder (Many Body Localization).
- **Statistical Mechanics**, mostly revolving around the non-equilibrium dynamics of quantum systems and the emergence of hydrodynamics.
- Universal features of **Quantum Information and entanglement** in non-equilibrium closed quantum systems.
- Dynamical phases, phase transitions and criticality in many body quantum systems from an information theoretic lens.

PUBLICATIONS/PREPRINTS

[2026] R. Thakur and S. Roy, **Dynamics of local quantum information in random unitary circuits**, [preprint] **arXiv:**2608.27440 doi.org/10.48550/arXiv.2608.27440

[2026] B. Pain, R. Thakur, and S. Roy, **Imprints of information scrambling on eigenstates of a quantum chaotic system**, Phys. Rev. B **114**, L140302, doi.org/10.1103/r1gj-g99c

[2025] R. Thakur, B. Pain, and S. Roy, **Logarithmic entanglement light cone from eigenstate correlations in the many-body localized phase**, Phys. Rev. B **111**, 174206, doi.org/10.1103/PhysRevB.111.174206

TALKS/SEMINARS

- **Scale resolved structure of quantum information under random quantum dynamics** at ICTS In-House Symposium, 2026.
- **Logarithmic entanglement lightcone from eigenstate correlations in the many-body localized phase** at ICTS In-House Symposium, 2025. Link: youtube.com/watch?v=2uMyxfGWoYM

POSTER PRESENTATIONS

- **"Dynamics of local quantum information in random unitary circuits"**, presented at the **International Workshop on Quantum many-body systems out of equilibrium: Relaxation, thermalization, and ergodicity breaking (2026)**, held at the Max Planck Institute for the Physics of Complex Systems in Dresden, Germany.
  Poster link: click here

PAST RESEARCH PROJECTS (NOT PUBLISHED)

- **Emergence of Hydrodynamics in classical circuit models** (2025)
  For more details `click here`
- **Constraints for Quantum Stability of the Inner Horizon of a Rotating Black Hole** (2024)
  For more details `click here`

Experience

WORK

2023-Present	**Research Scholar** International Centre for theoretical Sciences · Bengaluru
2022	**Research Intern** (S.N. Bhatt Memorial Excellence Fellowship) International Centre for theoretical Sciences · Bengaluru
2022-2023	**Quantum Networking Research Associate** QuLabs Software India · Remote, Part time
2021–2022	**Research Intern** IISER Tirupati · Remote

TEACHING

2026 | **Teaching assistant (Statistical Mechanics II)**
International Centre for theoretical Sciences · Bengaluru

VOLUNTEERING AND EXTRA-CURRICULAR EXPERIENCE

2020–2023 **Physics and Astronomy Club, IIT Roorkee**
Executive Member, Additional Secretary (2022–23)

2019–2023 **Music Section, IIT Roorkee**
Guitarist, Joint Secretary (2021-22)

2022 **Student Mentorship Program, IIT Roorkee**
Mentor to five freshmen at IIT Roorkee

Notable achievements and Awards

2022 **National Top 25**
National Graduate Physics Examination (NGPE-IAPT)

2016–2017 **State (Zonal) Rank: 10**
National Science Olympiad, Science Olympiad Foundation